PART A – THEORY QUESTIONS: IP Addressing

Below are simple answers suitable for beginners preparing for Linux, Networking, and DevOps interviews.

1. What is an IP Address?

An IP (Internet Protocol) Address is a unique logical address assigned to a device connected to a network. It is used to identify the device and communicate with other devices.

Example: 192.168.1.10

2. Why does every device require an IP Address?

Every device requires an IP address to identify itself on a network and communicate with other devices.

Example: A computer needs an IP address to access websites and communicate with servers.

3. What is IPv4?

IPv4 (Internet Protocol Version 4) is the fourth version of the Internet Protocol used to identify devices on a network.

Example: 192.168.1.1

4. How many bits are present in an IPv4 address?

An IPv4 address contains 32 bits.

It is divided into four octets, with 8 bits in each octet.

Example: 192.168.1.10

5. What are Octets in an IPv4 address?

An octet is a group of 8 bits in an IPv4 address.

An IPv4 address contains four octets separated by dots.

Example: 192 . 168 . 1 . 10

6. What is the valid range of each IPv4 octet?

The valid range of each IPv4 octet is 0 to 255.

- Minimum value: 0
- Maximum value: 255

Example: 192.168.1.10 is valid.

7. Differentiate between Public IP and Private IP.

Public IP	Private IP
Used to communicate over the internet.	Used within a local network.
Globally unique.	Can be reused in different private networks.
Assigned by ISP.	Usually assigned by router or DHCP server.
Accessible over the internet (subject to firewall and routing).	Not directly accessible over the public internet.

8. Give three examples of Private IP ranges.

There are three private IPv4 address ranges:

Class	Private IP Range
Class A	10.0.0.0 – 10.255.255.255
Class B	172.16.0.0 – 172.31.255.255
Class C	192.168.0.0 – 192.168.255.255

9. Who assigns a Public IP Address?

A public IP address is generally assigned by an Internet Service Provider (ISP).

Examples of ISPs in India:

- BSNL
- Airtel
- Jio

10. What is DHCP and why is it used?

DHCP stands for Dynamic Host Configuration Protocol.

It automatically assigns IP addresses and other network configuration details to devices.

It is used to avoid manually configuring IP addresses on every device.

11. What information does DHCP automatically provide to devices?

DHCP automatically provides:

1. IP Address
2. Subnet Mask
3. Default Gateway
4. DNS Server Address
5. Lease Time

12. What is a Default Gateway?

A Default Gateway is a device, usually a router, that allows devices in a local network to communicate with other networks, such as the internet.

Example: 192.168.1.1

13. What role does a Router play in a network?

A router connects different networks and forwards data packets between them.

Main functions:

- Connects LAN to the internet.
- Forwards data packets.
- Provides network routing.
- Often provides DHCP and NAT services.

14. Explain how a device accesses a website using ISP services.

When a device accesses a website, the following steps happen:

How a device accesses a website

1. User Device

User enters www.google.com

2. Router

Forwards the request to the ISP

3. ISP (Internet Service Provider)

Provides internet connectivity and routes traffic

4. DNS and Website Server

DNS finds the website's IP address, and the browser connects to the server

5. Website Opens

The server sends the response back to the device

Explanation:

1. The user enters a website address in the browser.

2. The device sends a request through the router.
3. The router forwards the request to the ISP.
4. DNS converts the domain name into an IP address.
5. The request reaches the website server.
6. The server sends the website data back to the device through the network.

15. Why is IP Addressing important in DevOps and Cloud environments?

IP addressing is important in DevOps and Cloud environments because it helps servers, applications, containers, and other devices communicate with each other.

Main uses:

1. Server communication – Connects different servers.
2. Cloud networking – Assigns IP addresses to cloud resources.
3. Load balancing – Distributes traffic between multiple servers.
4. Security – Controls network access using IP addresses and firewall rules.
5. Container networking – Allows Docker containers to communicate.
6. Kubernetes networking – Enables communication between Pods and Services.

Example: In AWS, EC2 instances receive private IP addresses for internal communication and can use public IP addresses for internet connectivity.